

Quantization of Chiral Gauge Theories in Two-Dimensions

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Abstract. It is shown that both abelian and non abelian chiral gauge theories in two dimensions can be made gauge invariant at the quantum level by adding a scalar field. In the bosonized form of the theory, the scalar field appears in a gauged Wess–Zumino action. The current algebra of the extended abelian theory is shown to be free of anomalous terms.

Quantum field theories with gauge couplings to chiral fermions suffer from the loss of gauge invariance. This “chiral anomaly” can be given a mathematically coherent description in terms of cocycles [1], which are objects appearing in the general representation theory of groups [2]. The anomaly arises due to the non-invariance of the fermionic determinant $D(A) = \det [\not{\partial} + A(1 - \gamma_5)]$. Under a gauge transformation $A \rightarrow A^g$, the determinant is multiplied by a phase [3]

$$D(A^g) = \exp[i2\pi\omega_1(A; g)]D(A) \quad (1)$$

where $\omega_1(A; g)$ is the 1-cocycle [1].

Recently it has been suggested [4] that a scalar field, h , with a gauged Wess–Zumino action (itself a 1-cocycle) should be added to a chiral gauge theory. The resulting quantum theory is then gauge-invariant due to the cocycle property

$$\omega_1(A^g; h) - \omega_1(A; gh) + \omega_1(A; g) = 0. \quad (2)$$

In the following, this proposal is examined in the context of abelian and non-abelian chiral gauge theories in two dimensions, where its main features can be explicitly demonstrated more easily. In both cases an auxiliary scalar field is added to a basic chiral gauge theory. Using the technique of bosonization via path integrals [5–8] the resulting theory is shown to be gauge invariant and contains a physical scalar field. The current algebra of the abelian theory is also

derived and is shown to have the non-anomalous behaviour appropriate for a consistent quantum theory.

Firstly, consider the Chiral Schwinger Model [9] to which a scalar field θ is added, resulting in a Lagrangian

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\bar{\psi}\gamma^\mu[\partial_\mu - ie\frac{1}{2}(1 - \gamma_5)A_\mu + ie\frac{1}{2}(1 - \gamma_5)\partial_\mu\theta]\psi. \quad (3)$$

Under a gauge transformation the various fields transform as

$$\begin{aligned} \psi &\rightarrow \exp[i\frac{1}{2}(1 - \gamma_5)A]\psi \\ A_\mu &\rightarrow A_\mu - (1/e)\partial_\mu A \\ \theta &\rightarrow \theta - A. \end{aligned} \quad (4)$$

The theory described by the Lagrangian (3) is, therefore, not gauge invariant at the classical level.

For the quantum theory the part of the generating functional involving θ is

$$Z_\theta = \mathcal{N} \int \mathcal{D}\theta \exp[-i \int d^2x \bar{\psi}\gamma^\mu \frac{1}{2}(1 - \gamma_5)\psi \partial_\mu\theta] \quad (5)$$

writing the chiral fermion current as

$$J^\mu = \bar{\psi}\gamma^\mu \frac{1}{2}(1 - \gamma_5)\psi$$

and performing a partial integration gives

$$\begin{aligned} Z_\theta &= \mathcal{N} \int \mathcal{D}\theta \exp[-i \int d^2x J^\mu \partial_\mu\theta] \\ &= \mathcal{N} \int \mathcal{D}\theta \exp[i \int d^2x \partial_\mu J^\mu \theta] \\ &= \mathcal{N}' \delta[\partial_\mu J^\mu]. \end{aligned} \quad (6)$$

So, introducing the auxiliary field θ is equivalent to imposing current conservation as a constraint on the theory at the quantum level.

In two dimensions the gauge field can be written as

$$A_\mu = -(1/e)(\partial_\mu\sigma + \varepsilon_{\mu\nu}\partial^\nu\rho). \quad (7)$$

Then the Lagrangian (3) can be written as

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\bar{\psi}\gamma^\mu[\partial_\mu - ie\frac{1}{2}(1 - \gamma_5)B_\mu]\psi \quad (8)$$

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where

$$B_\mu = -(1/e)[\partial_\mu(\sigma + \theta) + \varepsilon_{\mu\nu}\partial^\nu\rho]. \quad (9)$$

The fermions can then be decoupled by a chiral change of variables and integrated out [7, 8]. The Jacobian for this change is non-trivial [3] and must be regularized. The Dirac operator

$$\begin{aligned} \not{D} &= [\not{\partial} - \tfrac{1}{2}ie\tilde{B}] \\ \tilde{B}_\mu &= B_\mu^- + aB_\mu^+ \\ B_\mu^\pm &= (g_{\mu\nu} \pm \varepsilon_{\mu\nu})B^\nu \end{aligned} \quad (10)$$

is used as a regulator, where an arbitrary parameter a has been introduced. For $a = 0$ this corresponds to the Dirac operator appearing in (8). However, it should be noted that even for $a = 0$, $D[B_\mu]$ is not gauge covariant and hence the more general choice (10) is possible.

The Jacobian is easily found, by comparison with the result of [8], to be

$$\begin{aligned} \ln J_F &= (i/8\pi) \int d^2x [\rho \square \rho + \sigma \square \sigma - 2\sigma \square \rho \\ &\quad + a(\rho \square \rho - \sigma \square \sigma - \theta \square \theta - 2\theta \square \sigma) \\ &\quad + (2\theta \square \sigma + \theta \square \theta - 2\theta \square \rho)]. \end{aligned} \quad (11)$$

Under the gauge transformation (4) the various fields transform as

$$\begin{aligned} \sigma &\rightarrow \sigma + A \\ \theta &\rightarrow \theta - A \\ \rho &\rightarrow \rho \end{aligned} \quad (12)$$

and it can easily be shown that J_F is gauge-invariant for all values of a .

In this bosonized form the action contains a kinetic term for the scalar field θ which, therefore, represents a physical particle. To determine the particle spectrum the gauge is fixed by the choice

$$\sigma = 0 \quad (13)$$

leading to

$$\begin{aligned} \mathcal{L} &= \frac{1}{2e^2} \rho \square \square \rho + \frac{1}{8\pi} [(1+a)\rho \square \rho \\ &\quad + (1-a)\theta \square \theta - 2\theta \square \rho]. \end{aligned} \quad (14)$$

Introducing a field h by

$$h = \rho - (1-a)\theta \quad (15)$$

gives

$$\begin{aligned} \mathcal{L} &= \frac{1}{2e^2} \rho \square \square \rho + \frac{1}{8\pi} \left(\frac{a^2}{a-1} \right) \rho \square \rho \\ &\quad + \frac{1}{8\pi(a-1)} \partial_\mu h \partial^\mu h \end{aligned} \quad (16)$$

demonstrating the existence of one massive state and one massless state, exactly as in the original Chiral Schwinger Model [7–9].

Note that $a > 1$ is required, since for $a < 1$ the kinetic term for h has the wrong sign leading to a Hamiltonian

which is not positive semi-definite. ($a = 1$ is a special case [10]). For $a < 1$ the field h can be regarded as a ghost which restores gauge invariance.

Using the expressions

$$\rho = -e \square^{-1} \partial_\alpha \varepsilon^{\alpha\mu} A_\mu \quad (17)$$

$$\sigma = -e \square^{-1} \partial_\mu A^\mu$$

the Jacobian can be rewritten as

$$\begin{aligned} \ln J_F &= -(i/8\pi) \int d^2x [e^2 A_\mu (g^{\mu\alpha} + \varepsilon^{\mu\alpha}) \partial_\alpha \square^{-1} \\ &\quad \cdot \partial_\beta (g^{\beta\nu} - \varepsilon^{\beta\nu}) A_\nu \\ &\quad - a(\partial_\mu \theta - e A_\mu)^2 \\ &\quad + ([\partial_\mu \theta]^2 + e\theta[2\partial_\mu A^\mu - 2\varepsilon^{\mu\nu} \partial_\mu A_\nu])] \end{aligned} \quad (18)$$

which is the completely bosonized form equivalent to the Lagrangian in [11]

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_\alpha - \mathcal{L}_{\text{WZ}} \quad (19)$$

with the parameters of [11] given by

$$c_1 = \frac{e^2}{4\pi} \quad c_2 = -\frac{e^2}{4\pi} \quad \alpha = \frac{ae^2}{4\pi} \quad (20)$$

and $\mathcal{L}_{\text{WZ}}$ is a gauged Wess–Zumino action.

The current algebra of this theory can be derived from the fermionic generating functional

$$\begin{aligned} Z_F &= \int \mathcal{D}\psi \mathcal{D}\bar{\psi} \mathcal{D}\theta \exp \\ &\quad \cdot [i \int d^2x (\bar{\psi} \gamma^\mu [i\partial_\mu + e\frac{1}{2}(1-\gamma_5)B_\mu] \psi)] \end{aligned} \quad (21)$$

where the functional integration is over θ as well as ψ and $\bar{\psi}$. Then the quantities

$$\langle J_\mu \rangle = \frac{1}{Z_F} \frac{\delta Z_F}{\delta A^\mu} = \langle e\bar{\psi} \frac{1}{2}(1-\gamma_5)\psi \rangle \quad (22)$$

$$\langle J_\mu(x) J_\nu(y) \rangle = \frac{1}{Z_F} \frac{\delta^2 Z_F}{\delta A^\mu \delta A^\nu} = G_{\mu\nu}(x; y) \quad (23)$$

are the vacuum expectation values of the fermion currents in a classical background A_μ with the fermions and the θ -field quantized. Following Johnson and Low [12] current commutators can be calculated by

$$\begin{aligned} \langle [J_\mu(x), J_\nu(y)]_{x_0=y_0} \rangle &= \lim_{\epsilon \rightarrow 0} G_{\mu\nu}(x_1, x_0; y_1, x_0 + \epsilon) \\ &\quad - G_{\mu\nu}(x_1, x_0; y_1, x_0 - \epsilon). \end{aligned} \quad (24)$$

Using the Jacobian (11) the fermionic generating functional can be written as

$$\begin{aligned} Z_F &= \mathcal{N} \exp((i/8\pi) \int d^2x [(1+a)\rho \square \rho \\ &\quad + (1-a)\sigma \square \sigma - 2\sigma \square \rho]) F[\rho, \sigma] \end{aligned} \quad (25)$$

where

$$\begin{aligned} F[\rho, \sigma] &= \int \mathcal{D}\theta \exp(i/8\pi) \int d^2x [(1-a)\theta \square \theta \\ &\quad + 2(1-a)\theta \square \sigma - 2\theta \square \rho] \end{aligned} \quad (26)$$

since the functional integral $F[\rho, \sigma]$ is quadratic it can

be exactly evaluated, giving:

$$Z_F = \mathcal{N} \exp \left[\frac{-i}{8\pi} \left(\frac{a^2}{a-1} \right) \int d^2 x \rho \square \rho \right] \quad (27)$$

$$= \mathcal{N} \exp \left[\frac{ie^2}{8\pi} \left(\frac{a^2}{a-1} \right) \int d^2 x A_\mu (g^{\mu\nu} - \partial^\mu \square^{-1} \partial^\nu) A_\nu \right] \quad (28)$$

which coincides with the expression for the ordinary (non-chiral) Schwinger Model when $a = 2$. It is, therefore, obvious that the current commutators will have the form appropriate to a consistent quantum theory. An explicit calculation using (23), (24) and (28) yields

$$[J_0(x), J_1(y)]_{x_0=y_0} = \frac{-ie^2}{4\pi} \left(\frac{a^2}{a-1} \right) \delta(x-y) \quad (29)$$

$$[J_0(x), J_0(y)]_{x_0=y_0} = 0 \quad (30)$$

showing the non-anomalous behaviour of the $[J_0, J_0]$ commutator.

A non-abelian version of the Chiral Schwinger Model with an additional scalar field is represented by the Lagrangian

$$\mathcal{L}_g = \bar{\psi}(i\hat{\partial} + g^{-1}\hat{A}_-g + ig^{-1}\hat{\partial}_-g)\psi \quad (31)$$

where $\hat{A}_-$ and $\hat{\partial}_-$ are light-cone operators. In this case, however, the light-cone form arises because of the chiral coupling of the fermions (through the two dimensional relation $\gamma^\mu\gamma_5 = \gamma_\nu\epsilon^{\nu\mu}$) and not as a gauge choice.

Writing the gauge field A_- as

$$\hat{A}_- = -i(\hat{\partial}_-h)h^{-1} \quad (32)$$

then

$$g^{-1}\hat{A}_-g + g^{-1}\hat{\partial}_-g = -i(\hat{\partial}_-U)U^{-1} \quad (33)$$

where

$$U = g^{-1}h. \quad (34)$$

Under a gauge transformation the fields transform as

$$\begin{aligned} g &\rightarrow fg \\ h &\rightarrow fh \\ U &\rightarrow U \end{aligned} \quad (35)$$

so the classical action is not gauge invariant.

The fermions can be decoupled by a chiral rotation in the usual way, leading [13] to the bosonized form for the action

$$Z = \mathcal{N} \int \mathcal{D}g \mathcal{D}h \exp iW(g^{-1}h) \quad (36)$$

where

$$\begin{aligned} W(U) &= \frac{1}{2} \int d^2 x \text{tr} (\partial_\mu U \partial^\mu U^{-1}) \\ &\quad + \int d^2 x \int dr \epsilon^{\mu\nu} \text{tr} [(\partial_r U_r) U_r^{-1} (\partial_\mu U_r) U_r^{-1} (\partial_\nu U_r) U_r^{-1}] \end{aligned} \quad (37)$$

where r is a parameter which is integrated from 0 to 1 to build up a finite rotation from an infinitesimal one. This result is obtained by using the Dirac operator appearing in (31) as a regulator. This corresponds to $a = 0$ in the abelian case. The more general case can, of course, be handled but it will be sufficient here to consider only this specific case. Z is gauge invariant as a consequence of (35). As in the abelian case, the classical action is not gauge invariant but the quantum one is.

Including a kinetic term for the gauge bosons, the effective action is given by

$$S_{\text{eff}} = -\frac{1}{4} \int d^2 x F_{\mu\nu}^a(h) F_a^{\mu\nu}(h) + W(g^{-1}h) \quad (38)$$

which differs from the action of Rajaraman [14] by the addition of a term

$$\int d^2 x \int dr \epsilon^{\mu\nu} \text{tr} [(\partial_r h_r) h_r^{-1} (\partial_\mu h_r) h_r^{-1} (\partial_\nu h_r) h_r^{-1}]. \quad (39)$$

Thus it appears that Rajaraman's non-abelian action cannot be derived by bosonization from a chiral gauge theory. It nevertheless appears to be a consistent field theory [14].

In conclusion, it has been shown, by using the path integral method of bosonization, that adding a scalar field to the Chiral Schwinger Model renders the quantum field theory gauge invariant, despite the non-invariance of the classical Lagrangian. Upon bosonization the scalar field appears in a gauged Wess–Zumino action. The correct form of the Wess–Zumino term, needed to restore gauge invariance, is found by bosonization of the whole theory. Bosonization of just the part containing the scalar field would lead to an incorrect (non gauge invariant) result.

In this abelian theory it was shown that introducing the scalar field is equivalent to implementing gauge current conservation as a constraint on the theory at the quantum level. The current algebra for this model was also derived. It was shown that it has the non-anomalous form required for a consistent quantum theory.

A non-abelian version of the Chiral Schwinger Model was also considered. This theory can also be rendered gauge invariant at the quantum level by introducing a scalar field. Upon bosonization this scalar field appears in a non-abelian gauged Wess–Zumino action.

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Note Added. In a recent paper [15] the generating functional of a gauge theory is considered as an integral over the whole space of connections rather than as an integral over the orbit space. Where Weyl fermions are present the outcome is that the gauge transformations acquire the status of a physical field, which enters with a non-trivial action, the Wess–Zumino term being part of it. Thus it appears that the scalar fields considered above do not have to be added by hand, but arise naturally in the process of quantization.

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